3.4 Fundamental Constants and Conversion Factors

The numerical values of the fundamental constants used in our calculations are those obtained by Cohen and Taylor[8a] based on a least-squares adjustment of data existing in 1973. These values have been adopted by CODATA.[8a]

As mentioned in the Introduction (Sec.1), for tables issued since the Third Edition, values of the fundamental constants are based on the 1986 CODATA recommendations.[8b]

Fundamental constants

Quantity	Symbol	Value[a]
Speed of light in vacuum	c	299 792 458(1.2) m s^{-1}
Planck constant	h	6.626 176(36) × 10^{-34} J s
	$h/2\pi$	1.054 588 7(57) × 10^{-34} J s
Avogadro constant	N_A	6.022 045(31) × 10^{23} mol^{-1}
Elementary charge	e	1.602 189 2(46) × 10^{-19} C
Faraday constant, $N_A e$	F	9.648 456(27) × 10^4 C mol^{-1}
Absolute temperature of "ice point," O °C	T_O	273.150 0 K
Molar gas constant	R	8.314 41(26) J mol-1 K^{-1}
Boltzmann constant, RN_A^{-1}	k	1.380 662(44) × 10^{-23} J K^{-1}
Second radiation constant, hck^{-1}	c_2	1.438 786(45) × 10^{-2} m K

[a]Numbers in parentheses are the one standard deviation uncertainties in the last digits of the quoted value.

Conversion factors for energy-related units. 1973 CODATA constants

	kJ mol^{-1}	kcal mol^{-1}	eV (per molecule)	cm^{-1}	MHz
1 kJ mol^{-1}	1	0.239 005 7	0.010 364 35	83.593 47	2.506 069 × 10^6
1 kcal mol^{-1}	4.184	1	0.043 364 45	349.755 1	1.048 539 × 10^7
1 eV (per molecule)	96.4846	23.06036	1	8065.479	2.417 970 × 10^8
1 MHz	3.990 313 × 10^{-1}	9.537 077 × 10^{-8}	4.135 701 × 10^{-9}	3.335 641 × 10^{-5}	1
1 cm^{-1}	0.011 962 66	2.859 144 × 10^{-7}	1.239 852 × 10^{-4}	1	2.997 924 × 10^4
1 K (degree)	8.314 41 × 10^{-3}	1.987 19 × 10^{-7}	8.617 349 × 10^{-5}	0.695 030 4[a]	2.083 649 × 10^4
1 hartree	2625.500	627.509 5	27.211 606	2.194 746 × 10^5	6.579 684 × 10^9
1 rydberg	1312.750	313.754 7	13.605 803	1.097 373 × 10^5	3.289 842 × 10^9

[a]This value is inverse of the second radiation constant c_2; $(hc/k)^{-1} = c_2^{-1} = 0.695\ 030\ 4$.

3.5. Temperature Scale

Since the NIST-JANAF Thermochemical Tables deal with properties that vary with temperature, the temperature scale is of significance. The International Temperature Scales are devised to approximate, as closely as possible, the absolute thermodynamic (kelvin) temperature suggested by W. Thomson (Lord Kelvin) in 1854. The International Temperature Scale is established by the Comité International des Poids et Mesures (International Committee of Weights and Measures).

There have been four international temperature scales: The International Temperature Scale of 1927,[18] The International Temperature Scale of 1948 which was renamed the International Practical Temperature Scale of 1948 (IPTS-48) in 1960,[19–20] The International Practical Temperature Scale of 1968 (IPTS-68)[21] and The International Temperature Scale of 1990 (ITS-90).[22]

Our concern in the preparation of the NIST-JANAF Thermochemical Tables is to adjust any experimental measurements to the thermodynamic scale. In practice, this means converting to the ITS-90 scale since it is the currently accepted closest approximation to the thermodynamic scale.

To facilitate temperature adjustments, the article by Corruccini[19] was used for the 1927 to 1948 conversions and the article by Douglas[23] for the 1948 to 1968 conversion. Guildner and Edsinger[24] included a plot showing the deviation of IPTS-69 from the thermodynamic temperatures from 273.16 to 730 K. For conversions to the current scale (ITS-90), there are numerous articles which discuss the conversion of experimental data or smoothed results to the new temperature scale.

Ideal-gas tables of thermodynamic properties derived from statistical mechanics are based on the thermodynamic temperatures (as well as on the values of the physical constants used) and are hence independent of any practical temperature scale. The enthalpy of formation, Gibbs energy of formation, and logarithm of the equilibrium constant might depend on temperature-adjusted data.

For condensed phase tables, the temperature scale conversion is often not necessary and is not made since the correction would be small compared with the uncertainty in the data.